04 · Case 11: Pania's milk coming in

The words of this track, as it says them. Bold brown is the examiner.

Case 11. Lactation. Six marks.

Pania is two days postpartum, and her milk is just coming in. Two hormones are involved in the production and release of breast milk. For each one, say where it's made or secreted from, what stimulates it, and its function. Describe the structure of the mammary gland, and the path milk takes out of the breast. What type of gland is it, and what happens to the breasts in pregnancy as progesterone rises?

The second case she names for Test 3. It's long, but it's four small questions: the hormones, the gland, the milk's path, and pregnancy.

The hook: prolactin, pro for produces. Oxytocin lets it out. And the baby suckling fires both.

Hormone one: prolactin. It's secreted by the anterior pituitary. It stimulates milk production by the alveolar cells of the mammary glands.

Hormone two: oxytocin. It's made in the hypothalamus, and released from the posterior pituitary. It stimulates milk release, the let-down, by making the myoepithelial cells around the alveoli contract and squeeze the milk out.

What stimulates them? Both are released by a neural stimulus: the baby suckling at the breast. Let-down can also be set off by the baby's cry, its smell, or just thinking about the baby.

The gland. Mammary glands are modified sweat glands. Each breast has fifteen to twenty-five lobes, radiating around the nipple. Each lobe has smaller lobules, and the lobules hold the glandular alveoli, which make the milk. The pigmented areola surrounds the nipple, and suspensory ligaments attach the breast to the fascia of the chest muscle.

The milk's path out: alveoli, then alveolar ducts, then lactiferous ducts, then out at the nipple. Alveoli, alveolar ducts, lactiferous ducts, nipple.

And in pregnancy: rising progesterone develops the lobules and alveoli. Those are the milk factories. Oestrogen grows the ducts, the pipes. Before pregnancy the ducts are there, but not the glandular tissue that makes milk.

The traps. Swapping prolactin and oxytocin is the classic lost mark. Prolactin produces, oxytocin ejects. Second: the two halves of the pituitary. Prolactin comes from the anterior. Oxytocin is made in the hypothalamus and released from the posterior. Third: they're modified sweat glands, not sebaceous glands.

Once more. Pania's two hormones, the gland, the milk's path, and pregnancy.

Prolactin, from the anterior pituitary, stimulates the alveolar cells to produce milk. Oxytocin, made in the hypothalamus and released from the posterior pituitary, makes the myoepithelial

cells contract: the let-down. Both are triggered by the baby suckling. The mammary gland is a modified sweat gland: fifteen to twenty-five lobes around the nipple, lobules of milk-making alveoli, an areola, and suspensory ligaments. Milk goes alveoli, alveolar ducts, lactiferous ducts, nipple. In pregnancy, progesterone builds the lobules and alveoli, and oestrogen grows the ducts.